

LUKMAN EL HADI

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Seeking to join natixis for a 4-6 month internship in data science - 2026

Education

Ecole Centrale de Nantes

Engineering Degree - Data Science, Mathematics & Signal Processing

Sep 2022 – Sep 2025

Nantes, France

- Probability, statistics, stochastic processes.
- Optimization, inverse problems, Numerical methods, Signal processing, Corporate finance.
- Statistical modeling, Machine learning, Neural networks, Deep Learning, Image processing, Graph theory.

Université Paris 1 Panthéon-Sorbonne

Master 2 - Mathematics, Modeling and Methods in Economics and Finance (MMMEF - ex dea MMME)

Sep 2025 – Dec 2026

Paris, France

- Stochastic and Malliavin calculus, algorithmic trading.
- Financial products, derivatives pricing, yield curve and volatility modeling.
- Market risk, time series, statistical learning and Optimization.

Lydex - Lycée d'excellence de Benguerir

Preparatory Classes

Sep 2020 – Sep 2022

Benguerir, Maroc

- Intensive training for French and Moroccan competitive engineering entrance exams

Work Experience

Machine Learning Research Intern

LS2N (CNRS) / CHU Nantes - PET Image Reconstruction

Apr 2025 – Sep 2025 (6 months)

Nantes, France

- Reformulated **Positron Emission Tomography (PET)** image reconstruction as a **penalized maximum-likelihood inverse problem** under Poisson noise; Designed and trained **implicit neural representations** jointly with a physics-based forward operator (system matrix, sinogram projection).
- Developed a **stabilized input normalization scheme** that improved convergence robustness across varying image resolutions for the same dataset.
- Achieved up to **40%** faster convergence than baseline method and scored around **40% – 50%** higher reconstruction stability and performance.
- **First-author research paper** submitted to IEEE International Symposium on Biomedical Imaging (ISBI) 2026.

Private Tutor – Mathematics and Physics

Self-employed

Oct 2022 – Dec 2023 (1 year 3 months)

Nantes, France

- Provided one-on-one tutoring sessions to high school and undergraduate students in **mathematics** and **physics**.
- Monitored and mentored a group of students at **College Stendhal**, providing academic support in **mathematics** and **French**.

Projects

Probabilistic Electricity Price Modeling

Python / Energy Markets

Dec 2025 – Jan 2026

Université Paris 1

- Built a **probabilistic forecasting model** for day-ahead electricity prices, capturing **volatility clustering** and extreme price fluctuations.
- Modeled price uncertainty using a **Gaussian likelihood** framework, jointly estimating expected prices and conditional variance.
- Analyzed **price spikes and negative-price regimes** through dynamic confidence intervals and volatility diagnostics.
- Generated interpretable **risk indicators** relevant for portfolio exposure assessment and downside risk management.

Lookback Option Pricing under Heston Model

Python / Quantitative Finance

Dec 2025

Université Paris 1

- Developed a full **Monte Carlo pricing framework** for path-dependent exotic options (lookback call/put).
- Implemented **Heston stochastic volatility model** with correlated Brownian motions and full truncation scheme.
- Designed **log-Euler discretization** ensuring numerical stability and positivity of the underlying process.
- Estimated prices with **50,000 simulations** and constructed **confidence intervals** using CLT.
- Computed **Greeks via finite differences** (Delta, Vega, Rho, Theta) and analyzed sensitivity to model parameters.
- Conducted **convergence analysis** and studied impact of volatility-of-volatility and mean-reversion.
- Built a **risk analysis and hedging framework** (delta and vega hedging under stochastic volatility).

Achievements

- First-author paper accepted at GRETSI 2025.
- Ranked 5th out of 629 in the Moroccan national entrance exam to engineering schools.

Technical Skills

Programming: C++, Python (NumPy, pandas, SciPy, PyTorch,...), Matlab, Assembly (atmega328p)

Tools: Git, Linux, Docker, LaTeX

Languages: English, French, Czech, Arabic